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GENRE CLASSIFICATION FOR VIDEO COMPRESSION

Abstract

Systems and methods herein are for at least one execution unit that can perform an inference using a machine learning (ML) model and that is coupled to a video encoder, where the ML model can determine a genre associated with received frames of a media stream based in part on using ML model features associated with different genres, where the video encoder can encode the media stream based in part on the determined genre.

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Background/Summary

TECHNICAL FIELD

[0001] At least one embodiment pertains to video compression for frames of a media stream using an encoding that is based in part on a genre classification of one or more of the frames.

BACKGROUND

[0002] Video compression can be used to provide reduced media streams while preserving detail, to an extent, of content of an underlying video. Deep video compression techniques may be used with video compression. However, such deep video compression of a video may still require many parameters for tuning, to determine and limit operation of the video compression, for instance. A substantial part of the parameters provide different effects on different video. For example, one parameter may be used to improve a quality or to reduce a bitrate in part of a video being compressed. However, such one solution plan may have negative effects for different kind of content as the one solution may not suit the content under compression. While an approach may be to leave selection of parameters to users of the video compression, such as by an input to a configuration for the video compression, most users may not be informed about a relation between a video sequence of the content and available parameters to provide any benefit to the video compression. For example, a user may not be able to determine if there will be a positive or a negative impact using a parameter with a specific content. As a result, the parameters available may not be enabled by default and may not be used in a consumer environment for video compression.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0003] FIG. 1 is an illustration of a system for video compression using genre classification, in at least one embodiment;

[0004] FIG. 2 is an illustration of aspects of a machine learning (ML) model having sub-ML models to perform different inferences to provide genre classification for received frames, in at least one embodiment;

[0005] FIG. 3 is an illustration of aspects of a machine learning (ML) model having supervised training, unsupervised training, or semi-supervised training to provide genre classification for received frames, in at least one embodiment;

[0006] FIG. 4 illustrates computer and processor aspects of a system for video compression using genre classification, in at least one embodiment;

[0007] FIG. 5 illustrates a process flow for a system for video compression using genre classification, in at least one embodiment;

[0008] FIG. 6 illustrates yet another process flow for a system for video compression using genre classification, in at least one embodiment; and

[0009] FIG. 7 illustrates a further process flow for a system for video compression using genre classification, in at least one embodiment.

DETAILED DESCRIPTION

[0010] FIG. 1 is an illustration of a system **100** for video compression using genre classification, in at least one embodiment. The system **100** includes at least one circuit to perform as an encoder **104** that may be a video encoder and includes at least one other circuit to perform inference using a machine learning (ML) model **128**. For example, an inference by the ML model is to determine a genre in received frames of an input sequence **102**, for a media stream. As used herein, a genre may be represented by at least one feature that may discriminate different genre in an objective manner. Therefore, to determine a genre herein may be to determine, by an ML model, at least one feature for a genre. Further, as used herein, a genre may not be subjective as its underlying feature is objectively quantified and classified by an ML model. The genre, as used herein with respect to an ML model, may be different than readily apparent to a human observer. However, in at least one

embodiment, a feature may be tied to a genre and may be apparent to a human observer by the use of labeling, such as, in a supervised training of the ML model using labeled features tied to labeled genres.

[0011] Determination of a genre, by an ML model, is to enable the encoder **104** to perform efficient encoding that is suitable to the genre of an input sequence **102** in a media stream. As used herein, an input sequence **102** may include multiple frames subject to encoding after an underlying genre can be determined by a trained ML model **128**. In one example, the system **100** supports training of the ML model **128** to classify, into at least one genre, each input sequence **102** (or sets of scenes within the input sequences) of a media stream. Even though illustrated in the singular, an input sequence is continuously received and encoded. Therefore, the input sequence may include determination of different genres therein and may be subject to different encoding by different encoding parameters.

[0012] The system **100** is also enabled, using the encoder **104**, to choose default video compression parameters, reflecting the different encoding parameters, that are specific to the determined genre. The system **100** is also enabled, using the encoder **104**, to perform video compression or encoding that is devoid of some or all of the default video compression parameters. The video compression parameters are also referred to herein as encoding parameters. In at least one embodiment, the encoding parameters herein may include content-adaptive parameters and a configuration for total compression gain from the content-adaptive parameters. The total compression gain may be high relative to compression that is devoid of such content-adaptive parameters.

[0013] In at least one embodiment, the genre under classification herein may include, without limitations, natural content, camera help content, sports content, screen content, gaming content, cartoons, automotive content, medical imaging content, machine generated content, hand-held camera content, remote desktop applications, stationary camera content, and user-generated content (UGC). Different genres may benefit from different information retained during video compression. For example, medical imaging content may benefit from retaining information in areas pertaining to a study of a medical issue, whereas automotive content may benefit from retaining information in areas having automobiles in its surrounding areas. Therefore, any surrounding or remaining areas having content that is of less or no interest may be subject to higher compression than the retained information areas.

[0014] Genre classification by the ML model **128** enables the encoder **104** to perform encoding using encoding parameters that ensure retained information can maintain bits, for instance, in certain areas **102A** of interest, and that ensure compression to save bits in the surrounding or remaining areas **102B** of the content represented in the input sequence **102**. In at least one embodiment, instead of established genres, it is possible to explore different genres using specific features of video coding standards. For example, screen content for remote desktop applications may use specific encoding parameters that may be provided from an industry standards body. The use of such encoding parameters for training an ML model to associate against specific genres may be established using supervised learning, in at least one embodiment. In this manner, it is possible to objectify genres according to encoding parameters, other than noise features, different distributions of motion vectors, different intensity levels of pixels, or different edge features.

[0015] The training enabled for the ML model **128** may be performed naively by supervised learning, where tagging or labeling is provided in each sequence (or a set of scenes). Alternatively, the training enabled for the ML model **128** may be performed by unsupervised learning, where different video content (or areas thereof) may be initially subject to encoding sets using different encoding parameters, which may be classified by a trained ML model **128**. In yet another alternative, the training enabled for the ML model **128** may be performed by semi-supervised learning, where parts of the sequence may include a label and may be subject to classification along with other parts of the sequence, by the ML model **128**. In each approach, a best fit may be used by an ML model **128** to inform the training by establishing classes of different genres, for example.

Then, input video content of an input sequence **102**, during testing or in a live environment, may be classified against the established classes.

[0016] In one example, the ML model **128** may be trained using different features associated with the different genre. The different features may be associated with one or more of noise features, distributions of motion vectors, intensity levels of pixels, or edge features, in one instance. For example, one or more of the different noise features, the different distributions of motion vectors, the different intensity levels of pixels, or the different edges may be different for the different genres. For example, at least the intensity levels of pixels may be higher for certain part of the natural content versus certain medical imaging content. In an analogous manner, edges may be more defined for cartoon or screen content than for other types of genre.

[0017] In addition, the different noise features may include white noise in images, whereas the different distributions of motion vectors enables support for content of varying motion, such as a low median that may be the case in low relative motion within the content, against a high median that may be the case in high relative motion, than the low media, in the content. Further, the benefits of relying on intensity levels of pixels may be important to discriminate medical imaging content but not as important to discriminate natural content. Similarly, noise features and distributions of motion vectors may be different for gaming content and automotive content due in part to constant movement associated with such content. Therefore, such different features may discriminate different genres and may be used to define the different genre.

[0018] Once trained, the ML model **128** may use such features in an input sequence, which is to be encoded, to determine a genre in the input sequence by determining a best fit against a trained classes established by the ML model **128**. The encoder **104** may perform the encoding for the media stream that includes the input sequence **102**, based in part on encoding parameters that are suitable to the determined genre. The encoder **104** provides an encoded media stream, also referred to herein as an output bitstream. As detailed herein, however, the encoder **104** may be caused to provide different encoded media stream in a dynamic manner. For example, an initial input sequence may be encoded according to a first determined genre. Then, based at least in part on a scene cut event that introduces at least one further input sequence or a change in the input sequence (or set of scenes), a further classification by the trained ML model **128** may be performed.

[0019] For example, the initial genre may be determined by the ML model and informed to the encoder. However, the ML model may not perform inferences on further input sequence until the encoder informs the ML model of a scene cut event in the input sequence relative to a further input sequence. The dynamic encoding may be supported by a feedback loop from the encoder **104** to the ML model **128**. In at least one embodiment, one or more processors or execution units of a processor provide one or more different circuits that may be used to perform the encoder **104** distinctly from the ML model **128**. Based in part on the feedback in the feedback loop, a further inferred genre may be provided to the encoder to use different encoding parameters for the further input sequence. The ML model **128** may be trained to classify entire input sequences or a set of scenes into different genre so that the encoder can perform the encoding to include different encoding parameters for the different input sequences. However, the ML model **128** may be trained to classify each input sequence as it is received and based in part on a scene cut event to enable dynamic encoding by the encoder **104**.

[0020] Further, the classification by the ML model **128** may be used to enable a mode selection or parameter selection for an encoder **104** to perform encoding that is specific to the genre in an input sequence. For example, the mode selection or parameter selection may be associated with available ones of the encoding parameters to provide different but specific encoding for each genre or underlying feature. Further, while the ML model **128** may perform an initial classification or inference for a media stream, a scene cut event may be indicated back to the ML model **128** from the encoder **104** using the feedback loop **130**. The scene cut event may be associated with input sequences of the media stream, such as being within two different input sequences. The scene cut

event may be used as a point at which the ML model **128** is involved to determine a new genre of following scenes or a following input sequence.

[0021] In at least one embodiment, the use of the ML model **128** to classify input sequences of video content based in part on a scene cut event allows for dynamic encoding as genre changes occur in a media stream. An encoded media stream may, therefore, have different video sequences that correspond to the input sequences **102**, with different encoding parameters representing different genres as a result of different encoding required by changes in genre as a video content progresses and as testing and encoding occur in a dynamic manner. The different encoding parameters representing different genres may occur over time and are not required to be carried in the output bitstream **126** at a single point in time. However, it is also possible to provide the different encoding parameters representing different genres in the output bitstream **126** within a predetermined period if the inputs sequences have different genres determined by the ML model **128** or if the ML model **128** receives feedback and makes changes based in part on updates to the ML model **128** or different mode selection, for instance.

[0022] The training of the ML model may incorporate supervised training, unsupervised training, or semi-supervised training. In one or more of such training, at least a part, all, or none of any available scenes within an input sequence of different genres may be labeled. In at least the unsupervised and semi-supervised learning, a media stream of different genres may be part of an encoding set and the ML model classifies different features associated with different genres to cause an encoder to perform different encoding for the different genres in the input sequences. Therefore, the ML model enables a mode selection or parameter selection for an encoder that is suitable to a genre provided from the ML model. In addition, while described in the singular, the ML model may include sub-ML models, as described with one or more of FIGS. **2** and **3**, to perform different inferences for the input sequence. The inferences may cause the encoder to provide different encoding parameters in an output bitstream **126**.

[0023] In at least one embodiment, as genre changes occur for an input sequence **102**, it is also possible to provide the different inferences dynamically, which enable the different encoding parameters to also occur dynamically, over time for a video content associated with the output bitstream **126**. Separately, in response to the at least one encoding parameter indicated to the ML model **128** from feedback provided by the encoder **104**, it is possible to cause the encoder **104** to provide different encoding parameters in an output bitstream **126**. This may be to reflect adjustments or updates, by or within the ML model **128**. The adjustments or updates may be to associate at least one feature of a genre and an encoding parameter, based in part on the feedback. For example, an initial genre or associated feature may be indicated from the ML model **128** to the encoder **104**. However, the selection of encoding parameters itself is provided by the encoder **104**.

[0024] As the ML model **128** may not have the encoding information initially, the adjustments or updates after an initial genre or feature indicated to the ML model **128** may use sub-ML models to subsequently relate an encoding parameter with a feature for a genre. This process enables the sub-ML model to perform the adjustments or updates. Further, as the sub-models are of limited features or training, relative to a primary ML model, as described with respect to FIGS. **2** and **3**, the use of the sub-ML models enable size reduction to the ML model or speeding up of inferences by the ML model based on feedback dynamically received from the video encoder to the ML model.

[0025] In at least one embodiment, the encoder **104**, such as a video encoder, can receive at least one input sequence **102** associated with a media stream and can provide at least an output sequence **122** that is a compressed or changed version of the input sequence **102**. Further, the output bitstream **126** is an encoded media stream that includes different encoding parameters representing different genres and that corresponds to the input sequence **102**. For example, the different encoding parameters that are based on the different genres determined by the ML model **128**, in at least one embodiment. In a further example, different genres may be represented in the different encoding parameters by at least one value or parameter change of one encoding parameter of a set

of encoding parameters that represent the different encoding parameters. Therefore, different genres need not require all the different encoding parameter in the set to be different or have different values. As some video content may have a single genre all throughout, it is possible to make an initial determination of a genre that is used throughout the video content with the feedback only used to ensure that the encoding parameters are unchanged for the video content.

[0026] Further, it is also possible to cause updates to the encoding parameters if the feedback from the encoder to the ML model **128** indicates that a different feature is more prominent for the video content, even if the entire genre is unchanged. As such, the features may be separately trained to different sub-ML models and the feedback may be used to cause one of the sub-ML models to provide an inference for a different feature reflective of an aspect of a genre to be used as basis for encoding subsequent input sequences **102**. For example, as dynamic encoding is provided herein, with genre or feature changes that occur in a media stream, the output bitstream may be changed over time. Therefore, as used herein, a genre for the ML model **128** may pertain to one or more features that individually or altogether represent a genre of video content. As such, although the term genre is used herein, the genre and its classification, when described with reference to the ML model **128**, may be different than subjectively understood to a human observer. The genre and its classification, when described with reference to the ML model **128**, may be specific to specific features trained to the ML model **128** and may be different than subjectively understood to a human observer.

[0027] While illustrated in the singular, the encoding performed by the encoder **104** is to an input sequence or set of scenes that are all indicated as having the same genre, by the ML model **128**. The encoding performed is to provide an output bitstream that is an encoded media stream having different video sequences that are associated with different encoding parameters of the different genres, as determined by the ML model **128**. In at least one embodiment, the encoder **104** may be based in part on one of an H.264 standard, an MPEG2 standard, an AVC standard, an HEVC standard, a VP9 standard, an AV1 standard, or a VVC standard. However, the encoder **104** may be any encoder standard that allows weighting input, such as by mode selection using a quantization parameter (QP).

[0028] FIG. **1** illustrates that, in aspects of video encoding, a mode selection may be made to perform inter or intra mode coding. Such a mode selection may be performed using a mode selection module **116**. The mode selection may enable selection of parameters that may be associated with available ones of the encoding parameters. The result of such mode selection is to provide specific encoding based in part on the classification from the ML model **128**. The mode selection can also allow determination of how many bits the encoder **104** is willing to sacrifice in order to conceal and/or eliminate a distortion that may be relevant to certain parts of media content belonging to certain genres.

[0029] In at least one embodiment, there is trade-offs between bits used and distortion for the encoding performed. The trade-offs may be associated with distortion that may be different between different encoders. For example, the trade-offs may be between different user presets, different target bit rate (such as, possibly affecting a bit budget), and between different frames in a group of frames (GOP), representing an input sequence **102**, to be encoded. However, with the genre classification herein, the trade-offs may be suited to different genre so that useful information for a genre is preserved during encoding. In another example, a trade-off may include a possibility that some distortion occurs, within general areas **102B** of an input sequence **102**, and ensures that no distorting (or relatively lesser distortion) applies to certain areas **102A** of interest, during the encoding as pertinent to a genre determined for the input sequence.

[0030] Video compression may require intensive computation workloads with present state-of-the-art compression ratios providing 1/200 to 1/1000 compression but requiring more compute resources to perform such compression. However, with artificial intelligent and machine learning (AI/ML) workloads using large quantities of images and video, autonomous cars generating a large

amount of video in each car, applications like smart cities requiring more video data, content created for entertainment requiring higher video resolution and more bit depth, and present-day remote-working video conferencing technologies, it is appreciated that video compression must be performed in a more efficient manner. The efficient manner may be a reliance on genre determination by an ML model followed by an encoder to performing encoding using the determined genre. Further, it is appreciated that human eye limitations, along with the use of color space conversion and separation of luma (brightness) and chroma provide aggressive quantized or decimated features that may be limiting in providing quality video compression. Whereas, the use of an ML model enables objective genre determination for an encoder.

[0031] As used herein, a system **100** of an ML model **128** that is trained to classify genre in a video sequence, enables certain parameters or modes in an encoder **104** to encode an input sequence and to provide an encoded media stream that is also referred to as an output bitstream **126** herein. The encoding, supported by the ML model for genre classification, allows economy of bits. In one example, the ML model for genre classification enables the encoder to select areas in frames of input sequences to preserve quality therein by encoding these areas in specific encoding parameters to reduce an effect of the compression on the input sequences. This allows more bits, in encoding of areas **102A** of a video sequence, to achieve a desired quality where needed by a genre, and fewer bits to save on compute resources on other areas **102B** of the input sequences **102** where an associated genre does not require encoding of every detailed aspect therein.

[0032] As part of the encoding parameters, a Fourier or other related transform may be performed on blocks within every frame to convert data therein to a frequency domain and to allow quantization or discarding of information based on select frequencies. In doing so, transform coefficients at lower frequencies may be less aggressively quantized than those of higher frequency. Separately, motion estimation may be used to capture and encode movements across video frames. While all such approaches or options attempt to improve video compression, they may all serve a similar goal to allow an encoder to compress video into smaller bitstreams by eliminating noise, artifacts, allowing at least more intensive motion estimation and exploiting temporal and spatial redundancy. However, as used herein, for certain genres, additional benefits may be realized by retaining bits only to certain parts of a video sequence relevant to a genre. For example, the aggressiveness of transform and quantization, provided by a transformation and quantization (T and Q) module **108**, may be different for different genres.

[0033] In view of all such benefits, encoders may differ based in part on selections of proper tool(s) to enable aspects thereof to provide economy of bits. For example, the selections of proper tools is in reference to selection of encoding parameters to enable selection of areas (such as provided by macroblocks (MBs)) within frames of each input sequence **102** that are subject to more or less compression than other areas. This and other such approaches that may be defined within the encoder as different modes that may require more or less bits to ensure a desired quality. An RDO module **116A** may be associated with a mode selection module **116** of an encoder **104** to address requirements by the use of RDO metrics, such as Sum of Squared Errors (SSE) or Sum of Transformed Differences (SATD) to determine a cost associated with each selection made and to enable a selection based on the cost.

[0034] Further RDO metrics allow further mode selection that benefit from evaluation using further quality measures, including VMAF, SSIM, MS-SSIM, or PSNR. The addressing of temporal effect may remain for an encoder **104**, as it may be done on only a frame level using such further RDO metrics. Distortion may be determined as a difference from the original image. In at least one embodiment, the system **100** for video compression using genre classification herein includes the ML model **128** to enable improved selection of at least those quality measures that may be a basis for the mode selection provided by an RDO output **124** of the RDO module **116A**. The improved selection of at least the quality measures may be used by the encoder **104** to perform the video compression for video sequences **102** and, in particular, to provide the video compression that is

suitable to a genre. For example, the encoder **104** (also referred to herein as a video encoder) can receive transform coefficients or parameters, such as QPs. The RDO module **116A** operates to optimize, for each point or block of a frame, an efficient representation that may include segmentation, prediction modes, motion vectors (MVs), or the QPs.

[0035] In at least one embodiment, use of the RDO output **124** is to make a selection of a mode, as provided by the RDO module **116A**. The RDO also contributes to the encoding parameters available to be selected based in part on the genre classification for input sequences **102**. In at least one embodiment, an interface may be provided between the encoder **104** and the ML model **128** to allow input to be received in the ML model **128**, from the encoder **104**, that reflects feedback from a feedback loop **130**. Further, the interface can enable outputs to the encoder **104**, from the ML model **128**, which may be able to cause selection of certain video compression parameters for compression of input sequences **102**. The video compression parameters reflect quality measures of the RDO output **124**, for instance.

[0036] In at least one embodiment, an RDO may be limited to a single point for each block in each frame of an input sequence **102** and may be represented by a linear equation of $R + \lambda * D$, where **2** (λ) is a multiplier and where an (R, D) pair may be used with the multiplier to minimize a combined R+D value. R may be associated with a bit rate and D may be associated with distortion as it pertains to quality of the media. The RDO allows ranking, for instance, of candidate solutions using the linear equation to select one of the candidate solutions. Therefore, the λ value may be associated with a range from 1 to a minimized cost for the set of (R, D). R may be measured in bits and D may be a quality unit, such that the equation provides a measure of units of distortion for every bit of a bit rate used in a video compression process.

[0037] To achieve a predetermined bit rate of R, a certain value of λ may be used. The ML model **128** herein enables selection of encoding parameters that may include R, D, and λ values to allow the RDO to use different quality measures to different genre. This is performed to ensure that an effect of the video compression performed in the video encoder is based at least in part on the genre associated with the underlying video content. In at least one embodiment, therefore, the system **100** herein uses the ML model **128** to optimize an encoder **104** so that different quality measures, representing different video compression parameters, may be used with different genre.

[0038] In at least one embodiment, as illustrated in FIG. **1**, the encoder **104** is associated with at least one execution unit of a processor that performs inferences using the ML model **128**. The encoder **104** may include an output to provide feedback, through a feedback loop **130**, from the encoder **104** to the ML model **128**. In an example, the output can indicate a scene cut event or at least one of different encoding parameters used by or available in the video encoder to the at least one execution unit. For example, the at least one of the different encoding parameters are an encoding parameter of a prior input sequence.

[0039] The ML model **128** is able to use the at least one of the different encoding parameters of the prior video sequence to update the ML model **128** or to perform further training, retraining, or testing (including inferences) of the ML model **128**. Therefore, the ML model **128** can enable a different encoding to occur dynamically for the input sequence, such as for a subsequent input sequence that has a different genre, relative to a prior or initial input sequence, as determined by the ML model **128** and that is based at least in part on the scene cut event. The encoder **104** may also include an input to receive different inferences for the received frames of the subsequent input sequence, in response to the at least one of the different encoding parameters provided in the feedback loop. For example, the ML model **128** may include sub-ML models to provide the different inferences for subsequent input sequences, as described further with respect to FIGS. **2** and **3**.

[0040] In at least one embodiment, FIG. **1** provides an encoder **104** that is subject to H.264 encoding. The encoder **104** includes modules in hardware or software, such as a prediction module

112, the T and Q module **108**, and an entropy coding module **110**. There may be further modules, such as an inverse module **114**, a filter module **120**, a motion process module **118** (to support motion estimation and related aspects), and a prior or reference frames module **106**. The video compression using genre classification herein does not have effect on a decoding process for a bitstream provided from the encoder **104** that includes the output frame **122**. For example, the decoding process may be according to the H.264 decoding or other decoding relevant to the encoding format used to provide the output bitstream **126** from the encoder **104** and, particularly, as to the entropy coding module **110**.

[0041] A bitstream of frames, representing the input sequence **102** to be compressed may include different MBs. In at least one embodiment, different sizes of MBs may be supported in the encoder **104**, including but not limited to 8×8 , 8×16 , 16×8 , 4×4 , and 16×16 . The MBs likely correspond to displayed pixel data obtained at the location of the blocks. The prediction module **112** can generate a prediction MB that can be used to generate residual data reflective of data subject to quantization, as part of the video compression. There may be multiple prediction options associated with a prediction module **112**, including intra prediction that is associated with previously encoded data that is from a current sequence, such as the input sequence **102**. Another option associated with a prediction module **112** includes inter prediction that uses encoded data from other previously encoded frames, namely reference frames, such as from the prior or reference frames module **106**. These reference frames can appear before or after the current frame, in the display order and may be associated with motion compensation, such as motion process module **118** that uses previously coded frames, such as provided from the prior or reference frames module **106**.

[0042] Yet another option associated with a prediction module **112** includes the use of different prediction block sizes that is available to both, the intra prediction and inter prediction options. The use of different prediction block sizes of the MBs can change an accuracy associated with the predictions. A further option associated with a prediction module **112** includes the use of multiple frames during prediction, which is available in the inter prediction option to provide better accuracy in the predictions. A still further option is to skip MB data or residual data so that the encoder **104** itself performs an inference of the MB data based in part on the prediction MB. One or more of such options represent encoding parameters that may be applied to compress an input sequence **102** of a media stream based in part on a genre selection by an ML model **128**.

[0043] In at least one embodiment, intra prediction may be based at least in part on spatial data within at least one frame of an input sequence **102**. MBs generated as part of the intra prediction may be distinct from the MBs of the frame of the input sequence **102**. Residual data may be residual MBs generated by a subtraction of the prediction MB, from a current MB. The residual MB can be subject to transformation, quantization, and entropy coding in the provided modules **108**, **110** depending on a mode selected by a mode selection module **116** and that may be associated with the RDO module **116A** to perform the RDO, for instance. Further, in the encoder **104**, quantized data may be re-scaled and inverse transformed in the inverse module **114**. An output of the inverse module **114** may be filtered and combined with the prediction MB in the prediction module **112**. Motion estimation from the motion process module **118** may be included. The result may be a reconstructed MB or decoded frames that is provided to the prior or reference frames module **106** for further predictions. In at least one embodiment, the use of one or more of inter prediction or intra prediction represent additional encoding parameters that may be applied to compress an input sequence **102** of a media stream based in part on a genre selection by an ML model **128**.

[0044] FIG. 2 is an illustration of aspects **200** of a machine learning (ML) model having sub-ML models **1-N 210** to perform different inferences to provide genre classification for received frames, in at least one embodiment. In one example, the ML model **128** may include sub-ML models **1-N 210** to perform different inferences for the input sequence. The different inferences may be in response to the at least one of the different encoding parameters indicated by feedback to the ML

model **128** through a feedback loop **130** from the encoder **104**. This enables size reduction to the ML model **128** or speeding up inferences by the ML model **128**, based on the feedback. Further, as the feedback is dynamically received from the video encoder **104** to the ML model **128**, it is possible to update the ML model **128** or to perform further training or testing of the ML model **128**. [0045] FIG. 2 also illustrates that ML model **128** may be performed on different processor infrastructure **260B** than the encoder **260A**. Further, the ML model **128** may be controlled by an application **250** for which or on behalf of which the encoding is performed. The application **250** may provide control input to indicate that the ML model **128** is to perform the inference for input sequence **202**. Separately, the video encoder **104** may be controlled by its respective processing infrastructure **260A** to perform the encoding of the media stream based in part on capabilities associated with the processing infrastructure. For example, the capabilities may pertain to encoding standards enabled for the encoder **104**, including H.264 standard, an MPEG2 standard, an AVC standard, an HEVC standard, a VP9 standard, an AV1 standard, or a VVC standard. Further, the application **250** and the processing infrastructure **260A**, **260B** may share memory **270**, which may be part of the system **100**, to enable the inference and to enable the encoding of the media stream. [0046] In at least one embodiment, it is possible to use feedback of a scene cut event and at least one of the different encoding parameters together. While a genre **212** may be initially determined and indicated by the ML model **128** to the encoder **104** for use by the encoder **104**, feedback of one or more of a scene cut event or at least one of the different encoding parameters may be provided to the ML model **128** to enable determination of further genres for at least one subsequent input sequence **102**. In one example, a genre **212** is indicated as one or more values or other parameters that may be used in the encoder to select certain ones of the encoding parameters. However, the certain ones of the encoding parameters may not be known to the ML model **128**. Therefore, with the feedback in the feedback loop **130**, it is possible to inform the ML model **128** as to the encoding parameters used with the genre indicated or generally available in the video encoder. For example, feedback in the feedback loop **130** may indicate a scene cut event to the at least one execution unit and, subsequently, where an encoded media stream output from the video encoder is enabled to include different encoding parameters that are provided dynamically for the encoded media stream based at least in part on the scene cut event.

[0047] Thereafter, it is possible to update the ML model **128** to improve accuracy by training (including retraining) of the ML model **128** that associate together different encoding parameters used initially, with the genre indicated initially to the encoder **104**. Further, this process enables an encoded media stream that is the output bitstream **126** to include the different encoding parameters of the different genres, as determined by the ML model **128** and that may be provided dynamically for the input media stream having the input sequences **102** illustrated. Further, as the process is performed in an on-going or dynamic manner, the encoded media stream that is the output bitstream **126** may be changed based at least in part on the scene cut event and may include different encoding parameters as the genre changes in the content of the input sequences **102**.

[0048] The ML model **128** may include a genre features dataset **204** to retain features of the different genres that may be used for a primary ML model **208**. In at least one embodiment, the primary ML model **208** performs a determination of an initial genre for an input sequence **102**. Thereafter, a further determination may be performed dynamically, using the primary ML model **208** or using one or more of the sub-ML models **1-N 210**, for subsequent ones of the input sequences. In at least one embodiment, therefore, the ML model **128** is defaulted to a primary ML model **208** comprised therein. Further, like the ML model **128** having the genre features dataset **204**, the sub-ML models **1-N 210** may be associated with its own dataset that may be a portion of the genre features dataset **204**. However, the primary ML model **208** may use or access all the features of the genre features dataset **204**. Further, while a dataset may retain features, it may only do so to enable training, retraining, or updating of any one of the primary ML model **208** and the sub-ML models **1-N 210**. In one example, the different encoding parameters provided via feedback

to the ML model **128** may be provided to the genre features dataset **204** to be used in the retraining or updating of the ML model **128**.

[0049] The ML model **128** include a feature normalization module **206** that preprocesses the features to provide normalized features or to enable classification of marginal features into new or established classes. Therefore, the genre features dataset **204** and the feature normalization module **206** may be used for both training and testing of the ML model **128**. Further, the primary ML model **208**, the sub-ML models **1-N 210**, the genre features dataset **204**, and the feature normalization module **206** may be performed by distinct circuits that may be memory, cache, buffers, processors, or execution units within the processors. Therefore, the retraining or updating for the ML model **128** may apply to primary ML model **208** or any of the sub-ML models **1-N 210**.

[0050] In at least one embodiment, instead of the input sequence **102**, a processed sequence **202** that may be a downsampled or filtered sequence of the input sequence **102**, may be provided for the encoder **104** and to the ML model **128** to allow for application of video compression using genre classification, as described all throughout herein. A processed sequence **202** may be such that a color format conversion is provided to the input sequence **102**, in one non-limiting example. In at least one embodiment, the processed sequence **202** may be such that certain aspects that correspond to features for use by the ML model **128** may be enhanced or suppressed to assist the testing of the ML model **128**. Therefore, the ML model **128** may receive the processed sequence **202**, whereas the encoder **104** receives the input sequence **102**. In this manner, it is possible to reduce the workload on the ML model **128** for classifying the input sequence **102** as the classification may be performed on a downsampled version in the processed sequence **202**, while the encoding is performed using the input sequence **102**, as provided.

[0051] FIG. **3** is an illustration of aspects **300** of a machine learning (ML) model having supervised training, unsupervised training, or semi-supervised training to provide genre classification for received frames. Initially, supervised learning allows the ML model **128** to receive a set of labeled training data and is trained to recognize patterns in that data. Supervised learning may be provided for one or more of the features in the genre features dataset **204** by labeling associated with such features. Any of a primary ML model or a sub-ML model using such labeled features and having labeled classes may be regarded as a supervised ML model **302**. Then, further labeled features from a feedback loop may be used for training, retraining, or updates for the supervised ML model **302**. Separately, unsupervised learning may be provided for one or more of the features in the genre features dataset **204** by using the features as they exist and by causing the ML model **128** to determine patterns in the features without labels or instructions. Unsupervised learning allows determination of genres in unsupervised classes using existing features, relative to the supervised learning of a supervised ML model **302**. As the features are implied as having associations to different genre, the unsupervised classes are presumed to provide different genre classification for encoding parameters to be used with an input sequence **102**. For example, input sequences having features that classify under a certain one of the unsupervised classes of a trained ML model will cause encoding parameters of that unsupervised class to be used to perform encoding of the input sequences. There may be no labeling provided for such unsupervised classes. Any of a primary ML model or a sub-ML model using such unlabeled features and having unlabeled and unsupervised classes may be regarded as an unsupervised ML model **304**. Then, encoding parameters received in a feedback loop and to be associated with the one or more of the features in the unsupervised classes allows for further training, retraining, or updates for the unsupervised ML model **304**.

[0052] Further, semi-supervised learning may be provided for one or more of the features in the genre features dataset **204** by using the features in a combination of supervised and unsupervised learning. As such, a relatively smaller number of features, than used in supervised learning, may be labeled in the manner of the supervised learning. A relatively large number of features may be used even though unlabeled, in the manner of the unsupervised learning. The intent is to allow broader classification than provided by supervised learning and to provide classes even while being devoid

of a large number of labeled features. Semi-supervised learning allows determination of genres in such combined classes using existing features to provide a semi-supervised ML model **306**. As the features are a combination of labeled and implied features, having associations to certain genre, the combined classes provide different genre classification for encoding parameters to be used with an input sequence **102**. For example, input sequences having features that classify under a certain one of the combined classes of a trained ML model will cause encoding parameters of that combined class to be used to perform encoding of the input sequences. There may be labeling provided for such combined classes based in part on the labeled features, for instance. Any of a primary ML model or a sub-ML model using such combined features, having both labeled and unlabeled features, and providing combined classes may be regarded as a semi-supervised ML model **306**. Then, encoding parameters received in a feedback loop and to be associated with the one or more of the features in the unsupervised classes allows for further training, retraining, or updates for the semi-supervised ML model **306**, as a labeled or unlabeled feature.

[0053] In at least one embodiment, in each of FIGS. **2** and **3**, the genre features dataset **204** includes features such as, noise features for different genres, distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, and different edge features for the different genres. Further, a processed sequence **202** may be enabled using a downsampled or filtered sequence of the input sequence **102** that highlights one or more of such features. This process may be useful in training or testing the ML model **128**. Further, each of the sub-ML models **1-N 210** in FIG. **2** may be trained to each of the features. Therefore, each of the sub-ML models **1-N 210** may have its own sub-genre features dataset **204** that is associated with only one of the features. As such, if a feature is determined, from a feedback, as being prominent in the encoding of an input sequence **102**, the sub-ML model corresponding to that feature may be used with subsequent input sequences. Therefore, the system **100** supports inference, using the ML model, that is performed on processed versions or sequences **202** of one or more of the received frames. Further, the system **100** may also support inference, using the ML model, that is performed using one or more sub-regions **102A** of one or more of the received frames.

[0054] In at least one embodiment, it is also possible to use the processed sequence **202** for features of one or more of the supervised, unsupervised, or semi-supervised ML model or sub-ML models **302-306**. As certain features may be better classified than other features, it may be beneficial to a type of learning that is suitable to the clarity in the classification process. As such, while the encoder **104** receives the input sequence **102**, the ML model **128** may receive different types of the processed sequence **202** depending on the sub-models **302-306** used, in one instance. As such, the sub-ML models in FIGS. **2** and **3**, representing an ML model **128**, can perform different inferences for the received frames in the input sequence **102**, in response to at least one of the different encoding parameters provided as feedback to the ML model **128** or that is.

[0055] Still further, the sub-ML models in FIGS. **2** and **3**, representing an ML model **128**, are of different associated memory or processing capacities. As one or more of the sub-models may be trained using downsampled features and other sub-models may be trained using whole features, there may be distinct genre features datasets that are of different sizes and retained in different memory capabilities. Further, the primary ML model **208** and at least some of the sub-ML models **1-N 210** having whole features may require more processing capacity than those sub-ML models using downsampled features. The at least one execution unit can perform the ML model **128** using the primary ML model **208** or one of the sub-ML models **1-N 210**, in response to at least one of the different encoding parameters indicated via feedback to the ML model **128**. Further, the at least one execution unit can perform the ML model **128** based in part on a threshold capacity of at least one of the different associated memory or processing capacities. In one example, once trained, the primary ML model **208** or the sub-ML models **1-N 210** may be stored in memory and may be loaded to at least one execution unit to perform an inference based in part on the different encoding parameters to be provided in the output bitstream **126**.

[0056] FIG. 4 illustrates computer and processor aspects **400** of a system for video compression using genre classification, in at least one embodiment. For example, each of the illustrated processors **402** may include one or more processing or execution units **408** that can perform any or all of the aspects of the system **100** for video compression using an encoder and using genre classification from an ML model. The system **100** may include an interface that may be between the encoder and the ML model to allow the feedback loop and the genre to be communicated between these two aspects of the system.

[0057] The processing or execution units **408** may include multiple circuits to support the aspects described herein for one or more of the encoder **104**, the ML model **128**, and the interface between these two aspects. In at least one embodiment, the processors **402** may include CPUs, GPUs, DPUs that may be associated with a multi-tenant environment to perform one or more of the encoder **104**, the ML model **128**, and the interface between these two aspects described herein. Further, the GPUs may be distinctly in distinct graphics/video cards **412**, relative to a DPU (represented by a network controller **434**) and a CPU represented by the processors **402** illustrated in FIG. 4. Therefore, even though described in the singular, the graphics/video card **412** may include multiple cards and may include multiple GPUs on each card.

[0058] The computer and processor aspects **400** may be performed by one or more processors **402** that include a system-on-a-chip (SOC) or some combination thereof formed with a processor that may include execution units to execute an instruction, according to at least one embodiment. In at least one embodiment, the computer and processor aspects **400** may include, without limitation, a component, such as a processor **402** to employ execution units **408** including logic to perform algorithms for process data, in accordance with present disclosure, such as in embodiment described herein. In at least one embodiment, the computer and processor aspects **400** may include processors, such as PENTIUM® Processor family, Xeon™, Itanium®, XScale™ and/or StrongARM™, Intel® Core™, or Intel® Nervana™ microprocessors available from Intel Corporation of Santa Clara, California, although other systems (including PCs having other microprocessors, engineering workstations, set-top boxes and like) may also be used. In at least one embodiment, the computer and processor aspects **400** may execute a version of WINDOWS operating system available from Microsoft Corporation of Redmond, Wash., although other operating systems (UNIX and Linux, for example), embedded software, and/or graphical user interfaces, may also be used.

[0059] Embodiments may be used in other devices such as handheld devices and embedded applications. Some examples of handheld devices include cellular phones, Internet Protocol devices, digital cameras, personal digital assistants (“PDAs”), and handheld PCs. In at least one embodiment, embedded applications may include a microcontroller, a digital signal processor (“DSP”), system on a chip, network computers (“NetPCs”), set-top boxes, network hubs, wide area network (“WAN”) switches, or any other system that may perform one or more instructions in accordance with at least one embodiment.

[0060] In at least one embodiment, the computer and processor aspects **400** may include, without limitation, a processor **402** that may include, without limitation, one or more execution units **408** to perform aspects according to techniques described with respect to at least one or more of FIGS. 1-3 and 5-7 herein. In at least one embodiment, the computer and processor aspects **400** is a single processor desktop or server system, but in another embodiment, the computer and processor aspects **400** may be a multiprocessor system.

[0061] In at least one embodiment, the processor **402** may include, without limitation, a complex instruction set computer (“CISC”) microprocessor, a reduced instruction set computing (“RISC”) microprocessor, a very long instruction word (“VLIW”) microprocessor, a processor implementing a combination of instruction sets, or any other processor device, such as a digital signal processor, for example. In at least one embodiment, a processor **402** may be coupled to a processor bus **410** that may transmit data signals between processors **402** and other components in computer and

processor aspects **400**.

[0062] In at least one embodiment, a processor **402** may include, without limitation, a Level 1 (“L1”) internal cache memory (“cache”) **404**. In at least one embodiment, a processor **402** may have a single internal cache or multiple levels of internal cache. In at least one embodiment, cache memory may reside external to a processor **402**. Other embodiments may also include a combination of both internal and external caches depending on particular implementation and needs. In at least one embodiment, a register file **406** may store different types of data in various registers including, without limitation, integer registers, floating point registers, status registers, and an instruction pointer register.

[0063] In at least one embodiment, an execution unit **408**, including, without limitation, logic to perform integer and floating point operations, also resides in a processor **402**. In at least one embodiment, a processor **402** may also include a microcode (“ucode”) read only memory (“ROM”) that stores microcode for certain macro instructions. In at least one embodiment, an execution unit **408** may include logic to handle a packed instruction set **409**.

[0064] In at least one embodiment, by including a packed instruction set **409** in an instruction set of a general-purpose processor, along with associated circuitry to execute instructions, operations used by many multimedia applications may be performed using packed data in a processor **402**. In at least one embodiment, many multimedia applications may be accelerated and executed more efficiently by using a full width of a processor's data bus for performing operations on packed data, which may eliminate a need to transfer smaller units of data across that processor's data bus to perform one or more operations one data element at a time.

[0065] In at least one embodiment, an execution unit **408** may also be used in microcontrollers, embedded processors, graphics devices, DSPs, and other types of logic circuits. In at least one embodiment, the computer and processor aspects **400** may include, without limitation, a memory **420**. In at least one embodiment, a memory **420** may be a Dynamic Random Access Memory (“DRAM”) device, a Static Random Access Memory (“SRAM”) device, a flash memory device, or another memory device. In at least one embodiment, a memory **420** may store instruction(s) **419** and/or data **421** represented by data signals that may be executed by a processor **402**.

[0066] In at least one embodiment, a system logic chip may be coupled to a processor bus **410** and a memory **420**. In at least one embodiment, a system logic chip may include, without limitation, a memory controller hub (“MCH”) **416**, and processors **402** may communicate with MCH **416** via processor bus **410**. In at least one embodiment, an MCH **416** may provide a high bandwidth memory path **418** to a memory **420** for instruction and data storage and for storage of graphics commands, data, and textures. In at least one embodiment, an MCH **416** may direct data signals between a processor **402**, a memory **420**, and other components in the computer and processor aspects **400** and to bridge data signals between a processor bus **410**, a memory **420**, and a system I/O interface **422**. In at least one embodiment, a system logic chip may provide a graphics port for coupling to a graphics controller. In at least one embodiment, an MCH **416** may be coupled to a memory **420** through a high bandwidth memory path **418** and a graphics/video card **412** may be coupled to an MCH **416** through an Accelerated Graphics Port (“AGP”) interconnect **414**. In at least one embodiment, the graphics/video card **412** may be coupled to one or more of the processors **402** via a PCIe interconnect standard. Similarly, a network controller **424** may also be coupled to one or more of the processors **402** via a PCIe interconnect standard.

[0067] In at least one embodiment, the computer and processor aspects **400** may use a system I/O interface **422** as a proprietary hub interface bus to couple an MCH **416** to an I/O controller hub (“ICH”) **430**. In at least one embodiment, an ICH **430** may provide direct connections to some I/O devices via a local I/O bus. In at least one embodiment, a local I/O bus may include, without limitation, a high-speed I/O bus for connecting peripherals to a memory **420**, a chipset, and processors **402**. Examples may include, without limitation, an audio controller **429**, a firmware hub (“flash BIOS”) **428**, a wireless transceiver **426**, a data storage **424**, a legacy I/O controller **423**

connecting user input and keyboard interface(s) **425**, a serial expansion port **427**, such as a Universal Serial Bus (“USB”) port, and a network controller **434**. In at least one embodiment, data storage **424** may comprise a hard disk drive, a floppy disk drive, a CD-ROM device, a flash memory device, or other mass storage device.

[0068] In at least one embodiment, FIG. **4** illustrates computer and processor aspects **400**, which includes interconnected hardware devices or “chips”, whereas in other embodiments, FIG. **4** may illustrate an exemplary SoC. In at least one embodiment, devices illustrated in FIG. **4** may be interconnected with proprietary interconnects, standardized interconnects (e.g., PCIe) or some combination thereof. In at least one embodiment, one or more components of the computer and processor aspects **400** that are interconnected using compute express link (CXL) interconnects.

[0069] Therefore, the at least one execution unit **408** may be a circuit of at least one processor **402** to be associated with a video encoder. The association may be such that the at least one execution unit **408** of at least one processor **402** can perform the video encoder. The association may be such that the at least one execution unit **408** of at least one processor **402** can load and run or execute instructions to perform the video encoder. However, the association may be such that the at least one execution unit **408** of at least one processor **402** may be hardwired to perform the video encoder.

[0070] Further, the at least one execution unit **408** may be a circuit of at least one processor **402** to be associated with an ML model. The association may be such that the at least one execution unit **408** of at least one processor **402** can perform the ML model. The association may be such that the at least one execution unit **408** of at least one processor **402** can load and run or execute instructions to perform the ML model. However, the association may be such that the at least one execution unit **408** of at least one processor **402** may be hardwired to perform the ML model. Further, to support datasets, there may be other circuits, including the cache **404** that may be associated with the execution unit **408**. To perform the ML model, however, the trained ML model may be loaded to the execution unit **408** and run or executed from therein. In addition, there may be a different execution unit that provides an interface between the execution units performing the ML model and the encoder.

[0071] The ML model can be used to determine a genre associated with received frames of a media stream based in part on imparted training to the ML model using features associated with different genres. The imparted training may be a supervised training, an unsupervised training, or a semi-supervised training. The at least one execution unit **408** of at least one processor **402** performing the encoder can also enable the encoder to encode the media stream based in part on the determined genre. An encoded media stream provided by the encoder may include different video sequences that are associated with different encoding parameters of the different genres, as determined by the ML model.

[0072] Further, the at least one execution unit **408** of the at least one processor **402** is so that the features used in the ML model performed therein include one or more of different noise features for the different genres, different distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, or different edge features for the different genres. For example, these features may be used for the imparted training and may be also used in the testing of the ML model to indicate to the encoder a genre of an input sequence.

[0073] The at least one execution unit **408** of the at least one processor **402** that performs the ML model may include an input to receive feedback from the at least one different execution unit **408** performing the video encoder. The feedback may be an indication of a scene cut event to the ML model. The scene cut event may be a basis for the ML model to perform an inference to determine a genre for a subsequent input sequence. The genre may be the same as a prior genre of a prior input sequence. The genre may include a different feature than the prior genre of the prior input sequence based in part on the feedback that may include an encoding parameter that was used by the encoder for the prior input sequence. Thereafter, based in part on the genre indicated to the

encoder, an output bitstream having the encoded media stream may be enabled with the different encoding parameters of the different genres, as determined by the ML model and that is provided dynamically based at least in part on the scene cut event.

[0074] The at least one execution unit **408** of the at least one processor **402** that performs the ML model may include an input to receive feedback from the video encoder to indicate at least one of the different encoding parameters to the at least one execution unit. The ML model may include the sub-ML models to perform different inferences for the received frames in the input sequence, in response to the at least one of the different encoding parameters. For example, there may be a need for adjustments or updates, by or within the ML model, to associate at least one feature of a genre and an encoding parameter, based in part on the feedback. For example, an initial genre or associated feature may be indicated from the ML model to the encoder. However, the selection of encoding parameters itself is provided by the encoder. As the ML model may not have this information initially, the adjustments or updates after an initial genre or feature indicated to the ML model may use sub-ML models to subsequently relate an encoding parameter with a feature for a genre. This process can enable size reduction to the ML model or speeding up of inferences by the ML model based on feedback dynamically received from the video encoder to the ML model.

[0075] At least one execution unit **408** may be a circuit of at least one processor **402** to be associated with a video encoder to encode a media stream based in part on a genre associated with a media stream as determined using an ML model. The genre determined from received frames of the media stream may be based in part on imparted training to the ML model using features associated with different genres. An output bitstream of the encoder may be an encoded media stream that may include different video sequences that are associated with different encoding parameters of the different genres, as determined by the ML model. As a video content may be continuously subject to encoding and transmission to a decoder, it is appreciated that the output bitstream may have the different video sequences over a period of time at not at any instant, in at least one embodiment.

[0076] Further, the at least one execution unit **408** of the at least one processor **402** to be associated with a video encoder can include an output to provide feedback from the video encoder to the different one execution unit performing the ML model. The output can indicate a scene cut event or at least one of the different encoding parameters to the ML model. The different encoding parameters may be provided dynamically for the video stream based at least in part on the scene cut event and may be provided over time for the video content. The at least one execution unit **408** to be associated with a video encoder also includes an input to receive different inferences for the received frames. The different inferences may be in response to the at least one of the different encoding parameters provided via a feedback loop to the ML model. The ML model may use its included sub-ML models to provide the different inferences.

[0077] In at least one embodiment, at least one execution unit **408** of at least one processor **402** can be used to train an ML model using features associated with different genres for media streams. The ML model, once trained, is to enable a video encoder to encode a media stream based in part on a genre determined by the ML model for the media stream. Further, the ML model, once trained, is to enable the video encoder to provide an encoded media stream that includes different video sequences that are associated with different encoding parameters of the different genres, as determined by the ML model.

[0078] FIG. 5 illustrates a process flow or method **500** for a system for video compression using genre classification, in at least one embodiment. The method **500** includes receiving **502** a media stream that may include the input sequences described throughout herein. The method **500** includes performing **504** inference using an ML model to determine a genre associated with received frames of a media stream. This may be based in part on imparted training to the ML model using features associated with different genres. A verification **506** may be performed for a genre determined by the performing **504** step. The verification may be to ensure that classification is achieved at a

determined threshold to allow an inference by the ML model. In at least one embodiment, the method **500** includes encoding **508** the media stream using the video encoder based in part on the determined genre. An encoded media stream may be provided **510** as part of the encoding, where the encoded media stream includes different video sequences that are associated with different encoding parameters of the different genres, as determined by the ML model.

[0079] The method **500** of FIG. **5** may include a further step or may include a sub-step in which the features include one or more of different noise features for the different genres, different distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, or different edge features for the different genres. The method **500** of FIG. **5** may include a further step or may include a sub-step in which the imparted training is a supervised training, an unsupervised training, or a semi-supervised training.

[0080] FIG. **6** illustrates yet another process flow or method **600** for a system for video compression using genre classification, in at least one embodiment. The method **600** of FIG. **6** may be used with the method **500** of FIG. **5**. For example, the method **600** includes enabling **602** a feedback loop from the video encoder to at least one execution unit performing the inferences using ML model, which may be in step **504** of FIG. **5**. The feedback loop may be enabled based in part on feedback sent from the video encoder, in one example. Therefore, although the feedback loop may physically exist, there may be no useful information provided there through until the feedback is provided to enable the performance of the ML model of step **504**. In at least one embodiment, however, the enabling **602** step may be performed for subsequent input sequences of the media stream of step **502**.

[0081] The method **600** includes determining or verifying **604** that feedback is received. For example, the ML model may be associated with an interface to expect certain types of information that may be predetermined types of information. When such types of information is received, it may be processed as feedback received under step **604**. The method **600** include determining **606** a scene cut event from feedback in the feedback loop. The method **600** includes enabling **608** different encoding to be provided dynamically for the media stream based at least in part on the scene cut event. For example, initial encoding parameters may be provided for the encoded media stream, followed by adjustments, updates, or entirely different encoding parameters of different genres for subsequent input sequences in the media stream.

[0082] In at least one embodiment, the method **600** may include determining **606**, separately or in addition to a scene cut event from feedback in the feedback loop, at least one of the different encoding parameters provided with the feedback. The method **600** includes the enabling **608** for the different encoding to be provided dynamically for the media stream based at least in part on the scene cut event and/or the at least one of the different encoding parameters. For example, initial encoding parameters may be provided for the encoded media stream, followed by adjustments, updates, or entirely different encoding parameters for subsequent input sequences in the media stream. However, in at least one embodiment, the enabling **608** step for the encoding to be provided dynamically may use sub-ML models of the ML model to perform different inferences for the received frames in response to the at least one of the different encoding parameters in the feedback.

[0083] The method **600** of FIG. **6** may include a further step or may include a sub-step in which the sub-ML models are of different associated memory or processing capacities. The method **600** may include a further step or may include a sub-step in which a threshold capacity of at least one of the different associated memory or processing capacities may be determined. Then, one of the sub-ML models may be used in the method **600** to perform the different inferences, in response to at least one of the different encoding parameters indicated to the ML model, and based in part on the threshold capacity.

[0084] FIG. **7** illustrates a further process flow or method **700** for a system for video compression using genre classification, in at least one embodiment. The method **700** of FIG. **7** may be used with the method **500** of FIG. **5** or the method **600** of FIG. **6**. For example, the method **700** may be

associated with the training imparted to the ML model used in step 504 of the method 500 in FIG. 5. The method 700 in FIG. 7 includes determining 702 features that are associated with different genres for different video content. As described all throughout herein, the features may include one or more of different noise features for the different genres, different distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, or different edge features of the different genres.

[0085] The method 700 in FIG. 7 includes determining or verifying 704 to provide one or more ML models. For example, the determining or verifying 704 may apply to creating sub-ML models or to causing adjustments or updates to an existing primary ML model or to the sub-ML models. For example, when the feedback indicates encoding parameters for a genre that may not be associated with a best fit from a prior classification, there may be a new class associated with a new sub-ML model that may be provided to offer a further classification. The method 700 in FIG. 7 includes training 706 an ML model that may include the sub-ML models and a default primary ML model using features associated with the different genres. The method 700 in FIG. 7 includes using 708 the ML model, once trained, to enable a video encoder to encode a media stream based in part on a genre determined by the ML model. This may be in support of steps 504-508 of FIG. 5. The method 700 also includes enabling 710 the video encoder to provide an encoded media stream having different encoding parameters of the different genres, as determined by the ML model, in the manner of step 510 of FIG. 5.

[0086] In the following description, numerous specific details are set forth to provide a more thorough understanding of at least one embodiment. However, it will be apparent to one skilled in the art that the inventive concepts may be practiced without one or more of these specific details.

[0087] Other variations are within spirit of present disclosure. Thus, while disclosed techniques are susceptible to various modifications and alternative constructions, certain illustrated embodiments thereof are shown in drawings and have been described above in detail. It should be understood, however, that there is no intention to limit disclosure to specific form or forms disclosed, but on contrary, intention is to cover all modifications, alternative constructions, and equivalents falling within spirit and scope of disclosure, as defined in appended claims.

[0088] Use of terms “a” and “an” and “the” and similar referents in context of describing disclosed embodiments (especially in context of following claims) are to be construed to cover both singular and plural, unless otherwise indicated herein or clearly contradicted by context, and not as a definition of a term. Terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (meaning “including, but not limited to,”) unless otherwise noted. “Connected,” when unmodified and referring to physical connections, is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within range, unless otherwise indicated herein and each separate value is incorporated into specification as if it were individually recited herein. In at least one embodiment, use of term “set” (e.g., “a set of items”) or “subset” unless otherwise noted or contradicted by context, is to be construed as a nonempty collection comprising one or more members. Further, unless otherwise noted or contradicted by context, term “subset” of a corresponding set does not necessarily denote a proper subset of corresponding set, but subset and corresponding set may be equal.

[0089] Conjunctive language, such as phrases of form “at least one of A, B, and C,” or “at least one of A, B and C,” unless specifically stated otherwise or otherwise clearly contradicted by context, is otherwise understood with context as used in general to present that an item, term, etc., may be either A or B or C, or any nonempty subset of set of A and B and C. For instance, in illustrative example of a set having three members, conjunctive phrases “at least one of A, B, and C” and “at least one of A, B and C” refer to any of following sets: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, {A, B, C}. Thus, such conjunctive language is not generally intended to imply that certain

embodiments require at least one of A, at least one of B and at least one of C each to be present. In addition, unless otherwise noted or contradicted by context, term “plurality” indicates a state of being plural (e.g., “a plurality of items” indicates multiple items). In at least one embodiment, number of items in a plurality is at least two, but can be more when so indicated either explicitly or by context. Further, unless stated otherwise or otherwise clear from context, phrase “based on” means “based at least in part on” and not “based solely on.”

[0090] Operations of processes described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. In at least one embodiment, a process such as those processes described herein (or variations and/or combinations thereof) is performed under control of one or more computer systems configured with executable instructions and is implemented as code (e.g., executable instructions, one or more computer programs or one or more applications) executing collectively on one or more processors, by hardware or combinations thereof. In at least one embodiment, code is stored on a computer-readable storage medium, for example, in form of a computer program comprising a plurality of instructions executable by one or more processors.

[0091] In at least one embodiment, a computer-readable storage medium is a non-transitory computer-readable storage medium that excludes transitory signals (e.g., a propagating transient electric or electromagnetic transmission) but includes non-transitory data storage circuitry (e.g., buffers, cache, and queues) within transceivers of transitory signals. In at least one embodiment, code (e.g., executable code or source code) is stored on a set of one or more non-transitory computer-readable storage media having stored thereon executable instructions (or other memory to store executable instructions) that, when executed (i.e., as a result of being executed) by one or more processors of a computer system, cause computer system to perform operations described herein. In at least one embodiment, set of non-transitory computer-readable storage media comprises multiple non-transitory computer-readable storage media and one or more of individual non-transitory storage media of multiple non-transitory computer-readable storage media lack all of code while multiple non-transitory computer-readable storage media collectively store all of code. In at least one embodiment, executable instructions are executed such that different instructions are executed by different processors—for example, a non-transitory computer-readable storage medium store instructions and a main central processing unit (“CPU”) executes some of instructions while a graphics processing unit (“GPU”) executes other instructions. In at least one embodiment, different components of a computer system have separate processors and different processors execute different subsets of instructions.

[0092] In at least one embodiment, an arithmetic logic unit is a set of combinational logic circuitry that takes one or more inputs to produce a result. In at least one embodiment, an arithmetic logic unit is used by a processor to implement mathematical operation such as addition, subtraction, or multiplication. In at least one embodiment, an arithmetic logic unit is used to implement logical operations such as logical AND/OR or XOR. In at least one embodiment, an arithmetic logic unit is stateless, and made from physical switching components such as semiconductor transistors arranged to form logical gates. In at least one embodiment, an arithmetic logic unit may operate internally as a stateful logic circuit with an associated clock. In at least one embodiment, an arithmetic logic unit may be constructed as an asynchronous logic circuit with an internal state not maintained in an associated register set. In at least one embodiment, an arithmetic logic unit is used by a processor to combine operands stored in one or more registers of the processor and produce an output that can be stored by the processor in another register or a memory location.

[0093] In at least one embodiment, as a result of processing an instruction retrieved by the processor, the processor presents one or more inputs or operands to an arithmetic logic unit, causing the arithmetic logic unit to produce a result based at least in part on an instruction code provided to inputs of the arithmetic logic unit. In at least one embodiment, the instruction codes provided by the processor to the ALU are based at least in part on the instruction executed by the

processor. In at least one embodiment combinational logic in the ALU processes the inputs and produces an output which is placed on a bus within the processor. In at least one embodiment, the processor selects a destination register, memory location, output device, or output storage location on the output bus so that clocking the processor causes the results produced by the ALU to be sent to the desired location.

[0094] Accordingly, in at least one embodiment, computer systems are configured to implement one or more services that singly or collectively perform operations of processes described herein and such computer systems are configured with applicable hardware and/or software that allow performance of operations. Further, a computer system that implements at least one embodiment of present disclosure is a single device and, in another embodiment, is a distributed computer system comprising multiple devices that operate differently such that distributed computer system performs operations described herein and such that a single device does not perform all operations.

[0095] Use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments of disclosure and does not pose a limitation on scope of disclosure unless otherwise claimed. No language in specification should be construed as indicating any non-claimed element as essential to practice of disclosure.

[0096] In description and claims, terms “coupled” and “connected,” along with their derivatives, may be used. It should be understood that these terms may be not intended as synonyms for each other. Rather, in particular examples, “connected” or “coupled” may be used to indicate that two or more elements are in direct or indirect physical or electrical contact with each other. “Coupled” may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

[0097] Unless specifically stated otherwise, it may be appreciated that throughout specification terms such as “processing,” “computing,” “calculating,” “determining,” or like, refer to action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as physical, such as electronic, quantities within computing system's registers and/or memories into other data similarly represented as physical quantities within computing system's memories, registers or other such information storage, transmission or display devices.

[0098] In a similar manner, term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory and transform that electronic data into other electronic data that may be stored in registers and/or memory. As non-limiting examples, “processor” may be a CPU or a GPU. A “computing platform” may comprise one or more processors. As used herein, “software” processes may include, for example, software and/or hardware entities that perform work over time, such as tasks, threads, and intelligent agents. Also, each process may refer to multiple processes, for carrying out instructions in sequence or in parallel, continuously or intermittently. In at least one embodiment, terms “system” and “method” are used herein interchangeably insofar as system may embody one or more methods and methods may be considered a system.

[0099] In present document, references may be made to obtaining, acquiring, receiving, or inputting analog or digital data into a subsystem, computer system, or computer-implemented machine. In at least one embodiment, process of obtaining, acquiring, receiving, or inputting analog and digital data can be accomplished in a variety of ways such as by receiving data as a parameter of a function call or a call to an application programming interface. In at least one embodiment, processes of obtaining, acquiring, receiving, or inputting analog or digital data can be accomplished by transferring data via a serial or parallel interface. In at least one embodiment, processes of obtaining, acquiring, receiving, or inputting analog or digital data can be accomplished by transferring data via a computer network from providing entity to acquiring entity. References may also be made to providing, outputting, transmitting, sending, or presenting analog or digital data. In at least one embodiment, processes of providing, outputting, transmitting, sending, or presenting

analog or digital data can be accomplished by transferring data as an input or output parameter of a function call, a parameter of an application programming interface or interprocess communication mechanism.

[0100] Although descriptions herein set forth example implementations of described techniques, other architectures may be used to implement described functionality, and are intended to be within scope of this disclosure. Furthermore, although specific distributions of responsibilities may be defined above for purposes of description, various functions and responsibilities might be distributed and divided in different ways, depending on circumstances.

[0101] Furthermore, although subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that subject matter claimed in appended claims is not necessarily limited to specific features or acts described. Rather, specific features and acts are disclosed as exemplary forms of implementing the claims.

Claims

1. A system comprising: at least one execution unit to perform inference using a machine learning (ML) model to determine a genre associated with received frames of a media stream based at least in part on using ML model features associated with different genres; and a video encoder to encode the media stream based at least in part on the determined genre.
2. The system of claim 1, wherein an encoded media stream output from the video encoder comprises at least two video sequences that are associated with different genres.
3. The system of claim 1, wherein an encoded media stream output from the video encoder comprises different video sequences that are associated with different encoding parameters representing different genres.
4. The system of claim 1, wherein the features comprise one or more of different noise features for the different genres, different distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, or different edge features for the different genres.
5. The system of claim 1, wherein the ML model is trained using supervised training, unsupervised training, or semi-supervised training.
6. The system of claim 1, further comprising a feedback loop from the video encoder to the at least one execution unit to indicate a scene cut event to the at least one execution unit, wherein an encoded media stream output from the video encoder comprises different encoding parameters that are provided dynamically for the encoded media stream based at least in part on the scene cut event.
7. The system of claim 1, further comprising a feedback loop from the video encoder to the at least one execution unit to indicate at least one of different encoding parameters used by the video encoder to the at least one execution unit, wherein the ML model is comprised of sub-ML models to perform different inferences for the received frames in response to the at least one of the different encoding parameters.
8. The system of claim 7, wherein the sub-ML models are of different associated memory or processing capacities, and wherein the at least one execution unit is to use one of the sub-ML models, in response to at least one of the different encoding parameters indicated to the ML model, based in part on a threshold capacity of at least one of the different associated memory or processing capacities.
9. The system of claim 1, wherein the inference using the ML model is performed on processed versions of one or more of the received frames.
10. The system of claim 1, wherein the inference using the ML model is performed using one or more sub-regions of one or more of the received frames.
11. The system of claim 1, wherein the ML model is controlled by an application to perform the inference based in part on an input from the application and wherein the video encoder is controlled

by a processing infrastructure to perform the encoding of the media stream based in part on capabilities associated with the processing infrastructure.

12. The system of claim 11, wherein the application and the processing infrastructure share memory of the system to enable the inference and to enable the encoding of the media stream.

13. At least one execution unit to be associated with a video encoder, to perform an inference using a machine learning (ML) model to determine a genre associated with received frames of a media stream based in part on using ML model features associated with different genres, and to enable the video encoder to encode the media stream based in part on the determined genre.

14. The at least one execution unit of claim 13, wherein the features comprise one or more of different noise features for the different genres, different distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, and different edge features for the different genres.

15. The at least one execution unit of claim 13, wherein the ML model is trained using supervised training, unsupervised training, or semi-supervised training.

16. The at least one execution unit of claim 13, further comprising an input to receive feedback from the video encoder, the feedback to indicate a scene cut event to the at least one execution unit, wherein an encoded media stream output from the video encoder comprises different encoding parameters that are provided dynamically for the encoded media stream based at least in part on the scene cut event.

17. The at least one execution unit of claim 13, further comprising an input to receive feedback from the video encoder to indicate at least one of different encoding parameters used by the video encoder to the at least one execution unit, wherein the ML model is comprised of sub-ML models to perform different inferences for the received frames in response to the at least one of the different encoding parameters.

18. A video encoder to encode a media stream based at least in part on a genre associated with a media stream as inferred using a machine learning (ML) model performed on at least one execution unit, the genre determined from received frames of the media stream based at least in part on using ML model features associated with different genres.

19. The video encoder of claim 18, further comprising: an output to provide feedback from the video encoder to the at least one execution unit, the output to indicate a scene cut event or at least one of different encoding parameters used by or available in the video encoder to the at least one execution unit, wherein the different encoding parameters are provided dynamically for the encoded media stream based at least in part on the scene cut event; and an input to receive different inferences for the received frames in response to the at least one of the different encoding parameters, wherein the ML model is comprised of sub-ML models to provide the different inferences.

20. At least one execution unit to train a machine learning (ML) model using features associated with different genres for media streams, wherein the ML model, once trained, is to enable a video encoder to encode a media stream based in part on a genre inferred by the ML model for the media stream, and is to enable the video encoder to provide an encoded media stream based in part on the determined genre inferred by the ML model.

21. The at least one execution unit of claim 20, wherein the features comprise one or more of different noise features for the different genres, different distributions of motion vectors for the different genres, and different intensity levels of pixels for the different genres, or different edge features for the different genres.

22. A method for a video encoder, the method comprising: performing a machine learning (ML) model to infer a genre associated with received frames of a media stream based at least in part on using ML model features associated with different genres; and encoding the media stream using the video encoder based at least in part on the determined genre.

23. The method of claim 22, wherein the features comprise one or more of different noise features

for the different genres, different distributions of motion vectors for the different genres, different intensity levels of pixels for the different genres, or different edge features for the different genres.

24. The method of claim 22, further comprising: enabling a feedback loop from the video encoder to at least one execution unit performing the ML model; and determining a scene cut event from feedback in the feedback loop, wherein the different encoding is provided dynamically for the media stream based at least in part on the scene cut event.

25. The method of claim 22, further comprising: enabling a feedback loop from the video encoder to at least one execution unit performing the ML model; determining at least one of different encoding parameters used by or available in the video encoder, and provided in the feedback loop to the at least one execution unit; and using sub-ML models of the ML model to perform different inferences for the received frames in response to the at least one of the different encoding parameters.

26. The method of claim 22, wherein the sub-ML models are of different associated memory or processing capacities, and wherein the method further comprises: determining a threshold capacity of at least one of the different associated memory or processing capacities; and using one of the sub-ML models to perform the different inferences, in response to at least one of different encoding parameters from the video encoder and indicated to the ML model, based in part on the threshold capacity.
